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Compilation and verification of the engagement scale for college science and technology innovation competitions based on the three-dimensional theoretical model

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ABSTRACT

This research is dedicated to the development and validation of a multidimensional scale that is applicable to the engagement of college students in science and technology innovation competitions, with the objective of systematically assessing students' engagement behaviours and attitudes therein. Based on the three-dimensional engagement theory, which encompasses cognitive engagement, affective engagement, and behavioural engagement, this research comprehensively scrutinises relevant domestic and international literature. In conjunction with expert interviews and pre-surveys, three core dimensions and 29 preliminary measurement items of the scale are ascertained. With data from 716 students across four undergraduate institutions, through exploratory factor analysis, confirmatory factor analysis, as well as reliability and validity tests, an in-depth examination of the scale's structural validity, content validity, and reliability is carried out. The results indicate that the final scale consists of 25 items, and the reliability and validity indices of each dimension fulfil the psychometrics criteria, exhibiting high internal consistency and discriminative capacity. This scale is capable of effectively gauging the multidimensional engagement level of students in science and technology innovation competitions, furnishing reliable data support for educational administrators and laying a theoretical foundation for subsequent competition design and student engagement incentives. This work has furnished a vital instrument for science and technology innovation education in colleges and universities and a practical evaluation modality for facilitating the enhancement of students' innovation capabilities and comprehensive qualities.

KEYWORDS

Science and technology innovation competitions; engagement scale; three-dimensional engagement theory; reliability and validity verification

Introduction

The acceleration of global innovation-driven development has promoted innovation and entrepreneurship education becoming one of the core topics in the reform of higher education (Liu, 2023; L. N. Wang, 2023; H. C. Wang et al., 2023; Yao & Zhang, 2023). As an important means for universities to cultivate students' innovative abilities,

interdisciplinary cooperation abilities and practical abilities, science and technology innovation competitions have played an increasingly important role in promoting the all-round development of students (Chen et al., 2022; X. Q. Li & Wang, 2022). According to 2023 National Undergraduate Competition Analysis Report released by the Ministry of Education of the People's Republic of China, it covers multiple fields such as innovation and entrepreneurship, disciplinary competitions, and scientific and technological innovation, including a total of 84 events such as the 'Internet +' Innovation and Entrepreneurship Competition, the National Undergraduate Mathematical Modeling Competition, and the Challenge-Cup (China Association of Higher Education, 2024). Similarly, case competitions are widely used in higher education to enhance students' analytical and presentation skills, with well-known examples including NIBS, John Molson, Harvard ICC, CBS competitions, and the Global Undergraduate Awards. By participating in science and technology innovation competitions, students can not only obtain opportunities to solve practical problems but also cultivate key skills such as teamwork, time management, and innovative thinking (Tan & Huang, 2024). Multiple studies have shown that science and technology innovation competitions can significantly improve students' competitiveness (Zhong et al., 2023), scientific research qualities and abilities (Hou & Tian, 2024). However, there is still a lack of in-depth exploration in the systematic research on students' engagement behaviours in competitions. Although the importance of science and technology innovation competitions has been widely recognised, researches on the tools and methods for evaluating students' performance in the competition process remains insufficient. Currently, most of the existing studies mainly focus on the establishment of the education system (Song & Zhu, 2023; H. M. Xie et al., 2015) and the impact of competition results on students (Cheng et al., 2020), including academic performance and the improvement of innovative abilities. Nevertheless, there is a lack of scientific quantitative tools regarding how students participate in competitions, how much time and energy they invest in the competition process, and what their motivations for participating in competitions are. It is indicated that more research and practice are needed to improve the evaluation system for students' competition engagement behaviours and their effects, which will not only help to improve the quality of competitions but also better support the growth and development of students.

Fredricks et al. (2004) proposed the Three-Dimensional Framework of Engagement that mainly encompasses three dimensions: Cognitive Engagement, Emotional Engagement, and Behavioural Engagement. In the fields of education and psychology, this framework is widely applied and has become an important theory for researching students' sense of engagement. Just as the research pointed out (Shang & Gu, 2020), it is a common approach for dividing engagement into dimensions. Meanwhile, its high citation rate in relevant academic studies both at home and abroad is emphasised (X. Li & Li, 2021), indicating that the framework is both of broad influence and significant role in promoting in-depth understanding and research of student engagement. Specially, behavioural engagement refers to the explicit behavioural manifestations of students in learning activities, including engagement in classroom activities, attendance, discipline compliance, engagement in classroom interactions, and completion of assignments. Owing to that, it is usually the overt behaviour of students in academic settings, behavioural engagement is the dimension that is the easiest to observe and measure. Emotional engagement refers to students' emotional responses

to the classroom, school, teachers, and classmates during the learning process. Specifically, emotional engagement includes students' interest in learning, positive emotions towards the classroom atmosphere, a sense of belonging to the school, and a sense of closeness to teachers and classmates. Cognitive engagement refers to the mental effort invested by students in the learning process, as well as their understanding of learning content and intrinsic motivation. Cognitive engagement involves the metacognitive strategies, deep learning strategies, and perseverance in challenging tasks demonstrated by students when solving complex problems. Students with a high level of cognitive engagement will actively pursue understanding, engage in high-level thinking tasks, and exhibit relatively high autonomy and a willingness to take on challenges.

The conceptualisation of engagement as a multidimensional construct are not independent, there remains interconnections and interactions among these three dimensions. Behavioural engagement is the explicit manifestation of the other two dimensions. For example, when students develop a strong interest in learning (emotional engagement), they are more likely to exhibit positive behavioural engagement, such as completing assignments on time and actively speaking up in class. Similarly, when students engage in in-depth thinking during the learning process (cognitive engagement), their level of behavioural engagement may increase accordingly. Moreover, the relationship between emotional engagement and cognitive engagement is particularly close. For instance, the positive emotional experiences in learning like interest and enthusiasm will prompt them to engage more actively in learning tasks, increasing the depth of cognitive engagement. A high level of cognitive engagement can simultaneously enhance students' confidence and interest, thereby improving their positive emotional experiences.

In the Fredricks et al. (2004) compilation of engagement framework, it has provided a multi-dimensional method for pedagogy, enabling a more comprehensive understanding of students' learning engagement. Especially when compared with the traditional research approaches that solely focus on behavioural manifestations, this theory can cover a broader range of engagement manifestations. Such a panoramic perspective not only helps teachers and researchers evaluate students' engagement status but also promotes the reflection on and improvement of educational practices. Kahu (2013) applies the Three-Dimensional Framework of Engagement to the context of higher education and discovered that the interactions among behavioural, emotional, and cognitive engagement influence college students' learning outcomes. It is emphasised that emotional engagement can often be manifested through behavioural engagement and, in turn, affect the depth of cognitive engagement. Appleton et al. (2006) utilises the Three-Dimensional Framework of Engagement to explore the academic engagement of middle school students, with particular attention paid to those students at academic risk. It is indicated that behavioural and emotional engagement are the key factors in reducing the dropout rate, while cognitive engagement is significantly associated with long-term academic achievements. M. T. Wang and Eccles (2012) conducts a multi-level longitudinal analysis of adolescents' academic engagement, demonstrating that cognitive engagement is an important predictor of students' academic success, while emotional and behavioural engagement are more susceptible to the influences of family and school environments, which provides longitudinal data support for understanding the interrelationships among the three dimensions of engagement.

Although numerous scales have indeed provided certain reference frameworks for the assessment of academic engagement, they have demonstrated problems in effectively capturing the unique engagement patterns required in the specific field of science and technology innovation competitions during the practical application process. In science and technology innovation competitions, the connotation of engagement is not limited to individual academic performance but also encompasses a series of important factors such as teamwork, non-mandatory engagement, and interdisciplinary communication. However, existing scales often overlook or fall short in these crucial dimensions. Specifically, scholars like Fredricks classified engagement into behavioural engagement, emotional engagement, and cognitive engagement – three types. Nevertheless, their research failed to fully consider the role of teamwork as a key factor and the potential impact of competition feedback on improving the quality of engagement. In science and technology innovation competitions, teamwork is of vital importance since it can not only enhance the communication and collaboration abilities among students but also significantly facilitate the formation of innovative thinking and problem-solving capabilities. Moreover, the HESES scale proposed by Zhoc et al. (2019) covers multiple dimensions including academic, emotional, and social engagement and pays attention to students' investment of psychological energy in activities, it also neglects the characteristics of technology usage and online interaction in the context of science and technology innovation competitions, according to which deficiency is particularly evident in specific competition scenarios. Therefore, it is urgent to find more appropriate methods for assessment to better reflect the real situation. A model based on students' engagement behaviours is put forward through the research on innovation and entrepreneurship education in colleges, revealing the importance of students' engagement behaviours as well as emotional and cognitive aspects in innovation and entrepreneurship education (Ma & Wang, 2021). Meanwhile, focusing on the final results or academic achievements, existing studies overly neglecting students' motivations during the engagement process and their coping strategies when facing challenges. At the same time, they also fail to fully consider the significant impact of team interactions on the engagement experience. Such a one-sided focus may lead to misunderstandings about the engagement situation and, consequently, affect the comprehensive understanding and recognition of students' actual learning and engagement behaviours.

Overall, the existing engagement scales have failed to comprehensively and accurately reflect the real engagement status of college students in science and technology innovation competitions. This current situation not only restricts the in-depth understanding of students' engagement behaviours but also presents new challenges and opportunities for subsequent research work. Consequently, to make up for the deficiencies of the existing scales, we proposed to develop an engagement scale applicable to science and technology innovation competitions, which should cover the multi-dimensional engagement behaviours in science and technology innovation competitions, including students' engagement motives, time and energy investment, teamwork, resource utilisation, competition feedback, and achievements, among other aspects. Through systematic measurement, the scale can not only provide specific data support for college administrators

regarding students' engagement in competitions but also offer a theoretical basis for subsequent competition design, student motivation stimulation, and competition management. In this way, it is hoped that the scale will contribute to better promoting the growth and development of students in science and technology innovation competitions, enabling them to accumulate more experience and knowledge during the engagement process and thus enhance the overall engagement effect.

Based on this, this study proposes and validates a student engagement scale for university innovation competitions, grounded in the behavioural – emotional – cognitive model. Given the diversity of competition formats, the scale emphasises common features of student engagement to ensure broader applicability. The central research question is: how can student engagement in innovation competitions be systematically measured based on the three-dimensional model?

Research methodology

Sample selection

Online questionnaires are disseminated through Wenjuanxing to four domestic undergraduate institutions, comprising three public and one private. A sum of 1,020 questionnaires are amassed, with 653 female students and 367 male students. All the subjects of our research are undergraduates, mainly sophomores, with a total of 799 students, accounting for 78.33%, the second group is juniors, with a total of 77 students, accounting for 7.55%, and the numbers of freshmen and seniors are the same, both being 72 students, accounting for 7.06%, respectively. Subsequent to screening, 304 questionnaires characterised by uniform option selection, excessively brief response duration, and other anomalies are expunged. Ultimately, 716 valid questionnaires are procured, yielding an efficacy rate of 70.20%. The participants encompassed student cohorts who had engaged in diverse forms of science and technology innovation contests, like the 'Internet +' Innovation and Entrepreneurship Competition, the National Undergraduate Intelligent Vehicle Tournament, and the Mathematical Modeling Contest, with the aim of capturing the engagement behaviours of students with heterogeneous backgrounds and contest experiences. We employ the commonly used split-half validation method in the field of psychometrics (Floyd & Widaman, 1995; Worthington & Whittaker, 2006). After randomly bisecting the data, sample 1 ($n_1 = 358$) is allocated for item screening and exploratory factor analysis, whereas sample 2 ($n_2 = 358$) is deployed for confirmatory factor analysis and the verification of reliability and validity.

Scale development

Literature analysis and determination of dimensional items

To conduct a theoretical exploration of current engagement scales, an analysis of extant literature was done. Considering the common dimensions of measurement tools and actual competition engagement, this study distilled three key dimensions: cognitive engagement, behavioural engagement as well as emotional engagement. Table 1 shows existing engagement's structural dimensions and the measurement tools, with strict adherence to academic citation norms for research integrity.

Table 1. Summary of engagement measurement tools and dimensions.

Author	Measurement tools	Dimensions
F. Zhao et al. (2021)	Student Classroom Learning engagement Scale under the Flipped Classroom Model	Attention Concentration, Teacher-Student Interaction, Student-Student Interaction, Interest in Classroom, Critical Thinking, Classroom Satisfaction
Maroco et al. (2016)	University Student Engagement Inventory (USEI)	Cognitive engagement, Behavioural engagement, Emotional engagement
W. Xie and Yin (2022)	University Student Online Course Engagement Scale	Cognitive engagement degree, Behavioural engagement degree, Emotional engagement degree, Interactive engagement degree
Gunuc and Kuzu (2015)	Student Engagement Scale	Value, Sense of belonging, Cognitive engagement, Peer relationship, Relationship with faculty and staff, Behavioural engagement
Zhoc et al. (2019)	Higher Education Student Engagement Scale (HESES)	Academic engagement, Cognitive engagement, Social engagement with peers, Social engagement with teachers, Emotional engagement
R. Li (2016)	University Student Online Learning Engagement Scale	Cognitive engagement, Behavioural engagement, Emotional engagement
Handelsman et al. (2005)	Student Course Engagement Questionnaire (SCEQ)	Skill engagement, Emotional engagement, Participatory/Interactive engagement, Performance engagement
Lin and Huang (2017)	Student Course Engagement Scale (SCES)	Skill engagement, Emotional engagement, Performance engagement, Interactive engagement, Attitudinal engagement
Mazer (2012)	Student Interest and Engagement Scales	Student interest, Student engagement
Dixon (2015)	Online Student Engagement Scale (OSE)	Skill, Emotional engagement, Participatory/Interactive, Performance
M. Wang et al. (2016)	Maths and Science Engagement Scale	Behavioural engagement, Emotional engagement, Cognitive engagement, Social engagement
Marcionetti and Zammiti (2023)	Italian Higher Education Student Engagement Scale (I-HESES)	Academic engagement, Psychological engagement, Social engagement

The specific definitions of the three dimensions are detailed as follows:

Cognitive engagement (CP)

It focuses on students' mental efforts and comprehension in science and technology innovation competitions, involving the careful design of competition strategies, the creative generation and effective implementation of novel ideas, and other cognitive pursuits. Students with a high level of cognitive engagement tend to be actively engaged in competition-related activities and engage in deep, reflective thinking, which leads to a more thorough understanding and acquisition of knowledge.

Behavioral engagement (BP)

This relates to the tangible and committed actions and efforts of students in such competitions. The time and energy they devote, as well as the quality and frequency of their interactions with team members are all included. Those who are possessed of a significant degree of behavioural engagement are usually enthusiastic about participating in competition activities and complete tasks with great care and *dedication*.

Emotional engagement (EP)

It essentially pertains to students' emotional responses and attitudinal tendencies during competitions, including the interests, motivational drive, and emotional reactions to competition results. Students with a high level of emotional engagement typically show intense enthusiasm for competition activities, are fully immersed, and can derive pleasure and a strong sense of fulfilment.

Expert interviews and scale revision

Based on the above research of dimensions of science and technology innovation competition engagement construction, we drew on the existing engagement scales to develop new scale items. To guarantee the rationality and scientific validity of the scale, five college instructors possessing extensive experience in guiding science and technology innovation competitions and specialists in educational psychology were interviewed. In accordance with the feedback proffered by the experts, we carried out content revisions and improvements on the items within the scale. The experts predominantly recommended the incorporation of more elaborate items pertaining to students' teamwork and emotional vacillations into the scale, concomitantly ensuring an adequate number of items for each dimension to enable a comprehensive appraisal. Subsequently, preceding the formal administration of the survey, we carried out a pre-survey involving 50 students from two institutions of higher education to assess the lucidity of expression and manipulability of the questionnaire. In light of the outcomes of the pre-survey, certain items were refined to ensure their comprehensibility and capacity to precisely mirror students' engagement behaviours and affective experiences in the competitions. Ultimately, a preliminary scale comprising 29 items was established. The dimensions of the scale remained unaltered subsequent to the revisions. The scale adopted the Likert Scale, where 1 to 5 points successively denote 'strongly disagree', 'disagree', 'neutral', 'agree', and 'strongly agree'. The higher the total score, the higher the level of engagement in science and technology innovation competitions.

Statistical methods

For the data results of 358 subjects within Sample 1, we adopted SPSS 26.0 to conduct descriptive statistical procedures, correlation analyses, and exploratory factor analyses. In addition, we also utilised Amos 24.0 to conduct confirmatory factor analysis and tests of reliability and validity on the data results of 358 individuals in Sample 2.

Results

Item analysis

In the initial stage, the scores of the participants on the scale are systematically ranked in accordance with the overall scale score, the top 27% and the bottom 27% of the total test scores were, respectively, demarcated as the high-score group and the low-score group, with the remaining portion being designated as the intermediate group.

Sample 1 consists of 358 participants in total. Such scores were ranked from high to low, those with a ranking of 97 (with a score of 128 or above) and above were regarded as the high-score group, and those with a ranking of 262 (with a score of 104 or below) and below were considered as the low-score group. The remaining participants with scores in between were assigned to the intermediate group. Subsequently, we also conducted an independent-samples t-test, and it was found that there were significant differences between the high-score group and the low-score group for each item ($p < 0.001$), indicating that there exists no items that fail to show significant differences between the high-score group and the low-score group.

From the results, we found that there existed a highly significant difference between the high-score group and the low-score group on the scale. The t-value reached as high as 46.667, with a p-value less than .001, and the effect size was also remarkably large (Cohen's $d = 6.701$). The scale demonstrates excellent discriminatory power, suggesting that the differences are of practical significance. In particular, both Cohen's d and the Hedges' corrected d revealed a very strong effect, which implies that all items should be retained. The above results also demonstrate that the scale is capable of effectively distinguishing the discriminatory power between the high-score group and the low-score group in the test.

After the analysis of the results, we further calculated the correlations between the items and the total score. Firstly, the items with a correlation coefficient less than .40 were removed. It was found that the Cronbach's α coefficient of the processed scale did not increase. Since the correlation coefficient between item EP6 (I feel recognised in academic competitions) and the total score was .356, which was less than .4, it was thus deleted. The correlation coefficients of the remaining 28 items with the total score ranged from .714 to .864 ($p < .001$), the 28 items were therefore retained.

Exploratory factor analysis

We conducted an exploratory factor analysis on the 28 items based on the item analysis. The results demonstrated that the KMO value of the scale was as high as .961, far exceeding the recommended criterion of .6, which indicated a high degree of correlation among the items encompassed by the scale and its suitability for factor analysis. Additionally, the chi-square value of Bartlett's test of sphericity was significant ($\chi^2 = 9092.10$, $df = 378$, $p < .001$), further validating that the correlations among the items were not random but stemmed from the underlying factor structure. These data indices jointly supported the rationality and scientific nature of using this scale for factor analysis and provided a solid statistical foundation for the subsequent confirmatory analysis.

Next, we resorted to the principal component analysis technique to extract factors and adopted the varimax approach for orthogonal rotation. Given that the theoretical dimensional framework of engagement in science and technology innovation competitions has been established within this research paper, a predetermined number of three factors were extracted during the exploratory factor analysis process. The outcomes are exhibited in Table 2, which clearly demonstrates that none of the factor loading of any item was

Table 2. Exploratory factor analysis.

	Factor 1	Factor 2	Factor 3
BP5	.842		
BP4	.832		
BP10	.832		
BP9	.798		
BP1	.797		
BP7	.787		
BP3	.784		
BP8	.772		
BP11	.763		
BP6	.744		
BP2	.607		
CP5		.828	
CP1		.806	
CP2		.792	
CP8		.792	
CP3		.784	
CP7		.780	
CP10		.777	
CP4		.768	
CP9		.726	
CP6		.723	
EP4			.774
EP3			.760
EP1			.741
EP5			.716
EP2			.685
EP7			.651
EP8			.606

Table 3. Reliability coefficients of the science and technology innovation competition engagement scale.

Dimension	Internal Consistency Coefficient	Split-half Reliability
BP	0.921	0.908
CP	0.954	0.928
EP	0.940	0.904
Total	0.969	0.964

beneath.4, and there no situation of cross-dimensional loading. The textual statements of the items presented in [Table 2](#) can be found in [Appendix](#).

Reliability analysis

We employed Sample 2 to conduct the Internal Consistency Coefficient test, Split-half Reliability, and Test-Retest Reliability of the science and technology innovation competition engagement scale. The results demonstrated that the internal consistency coefficients ranged from .921 to .954, and the split-half reliability was between .904 and .928, indicating a relatively positive internal consistency of the scale (see [Table 3](#)).

Table 4. Model fit indices of the teacher teaching efficacy scale.

Indicator	χ^2/df	CFI	IFI	TLI	NFI	RMSEA
	2.658	0.944	0.944	0.938	0.913	0.068

Validity analysis

Content validity

Content validity focuses on whether the items of a scale can adequately represent the content being measured. During the process of constructing the measurement dimensions and compiling the texts of each item, we thoroughly combed through the relevant literature. On the basis of referring to the mature scales that have been widely used and boasted good reliability and validity, and in combination with expert seminars, we sorted out the core elements of engagement in science and technology innovation competitions. Eventually, we systematically constructed the dimensions of engagement in science and technology innovation competitions, the corresponding items as well as the number of items. After the initial scale was completed, a team consisting of 10 members, including psychology professors, doctoral students, and master's students, was invited to form a question-constructing group. Based on the criteria such as the rationality of item constructs, clarity of expression, conciseness, and comprehensiveness, we conducted multiple detailed discussions and evaluations on the dimensions of engagement in the scale as well as each individual item. This was done to ensure that the items formulated could cover the content scope of engagement in science and technology innovation competitions. We separately calculated the CVR values of the evaluations made by the 10 experts and scholars on the 22 items of the initial scale to ensure that the scale had satisfactory content validity.

Construct validity

Firstly, we conducted a confirmatory factor analysis on the scale and judged the fitting situation between the model and the observed data through the model fit indices. After modification, since the factor loading of BP1, BP2 and BP3 were less than 0.6, they were removed in our research. We utilised AMOS24.0 to verify the model fitting situation. Among the indices, RMSEA was .068, CFI was .944, IFI was .944, TLI was .938, and NFI was .913. All these indices reached positive standards, indicating that the scale model had a good fit. The specific data are shown in [Table 4](#) and [Figure 1](#).

To address potential multicollinearity (e.g. covariances = .93), we examined VIF values, all of which were well below the threshold of 10 (or more conservatively, below 3) (Hair et al., 2013), with the highest being 2.09.

During the model estimation process, a covariance path was added between the error terms of BP4 and BP5, as these items are semantically linked through the constructs of academic responsibility and initiative. Following the guidance of Brown (2015) and Kline (2023), correlating error terms is acceptable when items are conceptually similar, in order to improve model fit without undermining construct validity.

Secondly, to more precisely evaluate the construct validity, we calculated the correlations among different dimensions and between each dimension and the total scale. The data are shown in [Table 5](#), where the correlation coefficients among the dimensions range

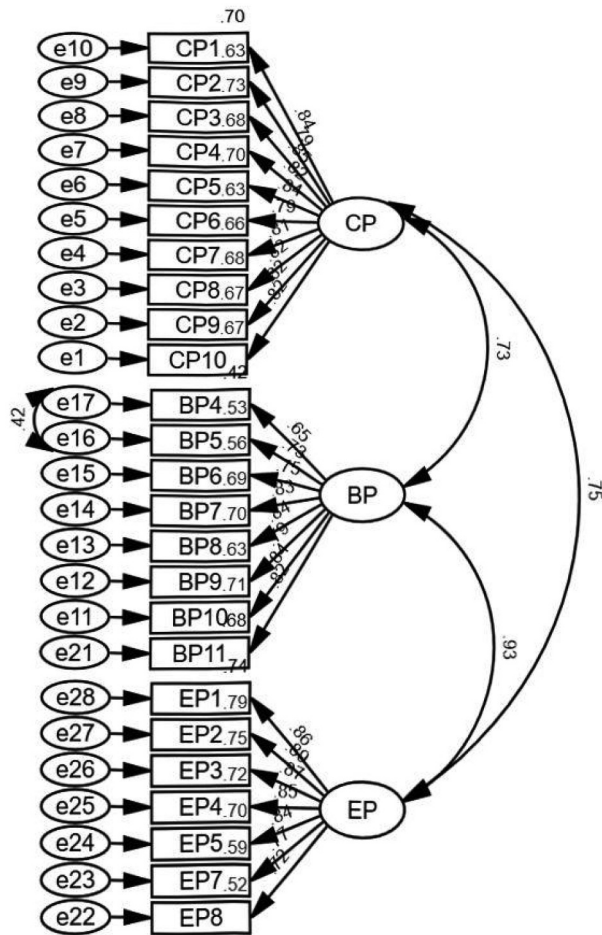


Figure 1. Confirmatory factor analysis.

Table 5. Correlation coefficients of the dimensions of the science and technology innovation competition engagement scale.

	BP	CP	EP
BP	1		
CP	.722**	1	
EP	.847**	.703**	1

* $p < 0.05$, ** $p < 0.01$.

Table 6. Scale convergent validity indicator.

	BP	CP	EP
AVE	.614	.689	.676
CR	.927	.939	.954

from .580 to .710, indicating a certain level of independence among the dimensions and the consistency of the content and concepts they measure with the total scale.

Finally as shown in Table 6, we computed the average variance extracted (AVE) of factor loading and the composite reliability (CR) is counted. According to the results, the AVE values of the cognitive engagement, behavioural engagement, and emotional engagement were close to .70, demonstrating sound convergent validity. Moreover, the CR values of the three dimensions were .927939, and .954 respectively, all exceeding .90, which implies that the designed scale has a high internal consistency and excellent reliability.

Discriminant validity was assessed using Pearson correlations between constructs. According to the criteria suggested by Kline (2016) and Fornell and Larcker (1981), correlations below 0.85 indicate acceptable discriminant validity. As shown in Table 5, all inter-construct correlations fall below this threshold, confirming adequate discriminant validity among the constructs.

Discussion

Drawing upon an exhaustive synthesis of extant domestic and international research, we constructed the Science and Technology Innovation Competition Engagement Scale, which subsumes three cardinal dimensions: cognitive engagement, behavioural engagement, and emotional engagement. Through a series of meticulous procedures including item analysis, exploratory factor analysis, and comprehensive reliability and validity examinations, we have ascertained that all the indices associated with the scale duly conform to the stringent benchmarks of psychometrics.

In the course of scale development, we have implemented a battery of rigorous psychometric protocols, such as item analysis, exploratory factor analysis, and confirmatory factor analysis. The above-mentioned steps guarantee that each entry of the scale can precisely mirror the engagement experience of college students in science and technology innovation competitions. The internal consistency reliability and test-retest reliability of the scale have both reached acceptable levels, which indicates that the scale possesses good reliability. In addition, the construct validity of the scale has been verified, the correlations among the various dimensions demonstrate the construct validity of the scale.

Notwithstanding the fact that the scale has manifested favourable reliability and validity within the current sample cohort, we are acutely aware that the external validity of the scale warrants further substantiation across a heterogeneous array of colleges and universities, varying in terms of type and scale. Future research endeavours could potentially explore the application of the scale to undergraduate students hailing from disparate cultural milieus, with the objective of evaluating its generalisability and applicability within a more expansive demographic spectrum. Moreover, the long-term validity of the scale, specifically its efficacy in prognostication of students' subsequent engagement behaviours in science and technology innovation competitions, represents a salient avenue for future research investigations.

The scale development and validation has provided a reliable instrument for educational researchers and practitioners to assess and understand the engagement motives and experiences of college students in science and technology innovation competitions.

Through further research and application, it is expected that the scale offers valuable insights and intervention strategies for enhancing college students' engagement. However, what deserves attention is that cognitive and emotional engagement are rather difficult to measure through external observation and usually rely on students' self-reports, leading to the possible social desirability bias in the process of scale compilation. Future research can further verify the validity of the scale by combining experimental designs with qualitative research methods.

Secondly, as students from different countries or regions have diverse motives, habits and cultural backgrounds when participating in innovation competitions, the applicability and universality of the scale may be challenged in cross-cultural studies. Finally, since science and technology innovation competitions involve a variety of different tasks and activities, behavioural, emotional and cognitive engagement may vary significantly at different competition stages. The compilation of the scale was carried out based on the common engagement characteristics of college students in science and technology innovation competitions. Future research could consider focusing on different types of competitions for college students, respectively, like innovation and entrepreneurship contests, discipline competitions and professional skills competitions, etc., to explore the heterogeneity under different competition types.

Simultaneously, Ma and Wang (2021) carried out an in-depth investigation into the domain of college innovation and entrepreneurship education. The findings unveiled that students' engagement in innovation and entrepreneurship education hinges not merely upon their individual interests and motivations but is also conspicuously affected by the supportive milieu proffered by educational institutions. Consequently, it is recommended that universities ought to optimise the allocation of educational resources with the aim of effectively augmenting students' engagement levels and thereby catalysing the advancement of their innovation and entrepreneurship capabilities.

In summary, the engagement scale developed in this study meets established psychometric standards, providing a reliable tool for assessing university students' participation in innovation competitions. It is important to note that the participants were primarily involved in in-person competitions, and thus, the current validation is most applicable to offline contexts. Future studies may further explore the scale's applicability in online or hybrid competition formats, and examine how the results can inform more targeted educational interventions to enhance student engagement across various competition settings.

Conclusions

- (1) We proposed the Science and Technology Innovation Competition Engagement Scale consisting of three dimensions: cognitive engagement, behavioural engagement, and emotional engagement, with a total of 25 items.
- (2) The designed scale demonstrates satisfactory reliability and validity, with a relatively high goodness – of – fit of the model, which can comply with the requirements of psychometrics and is capable of meeting the measurement needs of science and technology innovation competition engagement. Therefore, it can serve as an effective tool for evaluating the engagement degree of college students in science and technology innovation competitions.

- (3) Grounded in the three-dimensional engagement framework, this study addresses the core research question – how to systematically measure university students' engagement in innovation competitions – thus providing a theoretical and methodological foundation for future studies across different competition formats and participant profiles.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Notes on contributors

Quan Zheng, Ph.D., is a researcher at Sehan University in South Korea. His work focuses on college students' innovation and science and technology innovation competitions. He has successfully guided students to win several national competition awards.

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Guolai Wu, Ph.D., is a professor at the Faculty of Psychology at Tianjin Normal University and a professor at Sehan University's Graduate School. He is also an associate editor for the journal *Psychological and Behavioral Research*. His research areas include self-efficacy, father presence, implicit learning, mind wandering, and sustained attention.

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Appendix

College Students' Participation Scale in Scientific and Technological Innovation Competitions

Cognitive Engagement (CP)

- (1) I enjoy the intellectual challenges brought by academic competitions.
- (2) I gain a great sense of satisfaction from academic competitions.
- (3) I find academic competitions intellectually inspiring.
- (4) I am usually motivated to participate in academic competitions.
- (5) When preparing for academic competitions, I ask myself questions to ensure I understand the content.
- (6) Even when there is no competition, I actively learn knowledge related to the competition.
- (7) I review and correct my mistakes in the preparation for academic competitions.
- (8) During academic competition preparation, I often think about how the competition content connects with other knowledge.
- (9) I enjoy thinking about complex problems in academic competitions and trying to come up with solutions.
- (10) When preparing for academic competitions, I consider the real-world applications of the competition content.

Behavioral Engagement (BP)

- (1) I usually complete the preparation tasks for academic competitions on time.
- (2) When I have questions, I ask and participate in discussions related to academic competitions.
- (3) I am an active participant in academic competitions.
- (4) I voluntarily participate in academic competitions.
- (5) I review knowledge before the competition to ensure I understand the content.
- (6) I regularly attend academic competition training and participate in discussions.
- (7) I complete academic competition tasks on time.
- (8) I frequently communicate with teammates to share competition insights.

Emotional engagement (EP)

- (1) I feel happy when participating in academic competitions.
- (2) I feel excited and passionate about participating in academic competitions.
- (3) I enjoy participating in academic competitions.
- (4) Academic competitions meet my expectations.
- (5) I really like my academic competition team.
- (6) I feel a sense of accomplishment when participating in academic competitions.
- (7) I feel satisfied with academic competitions

Note. Items BP1–BP3 and EP6 were excluded from the final validated measurement model based on the results of the confirmatory factor analysis (CFA) and the correlation analysis with the total score. These items did not meet the required psychometric criteria for inclusion and were therefore removed from the final scale. Future researchers are advised to use only the retained items when applying the scale in empirical studies.